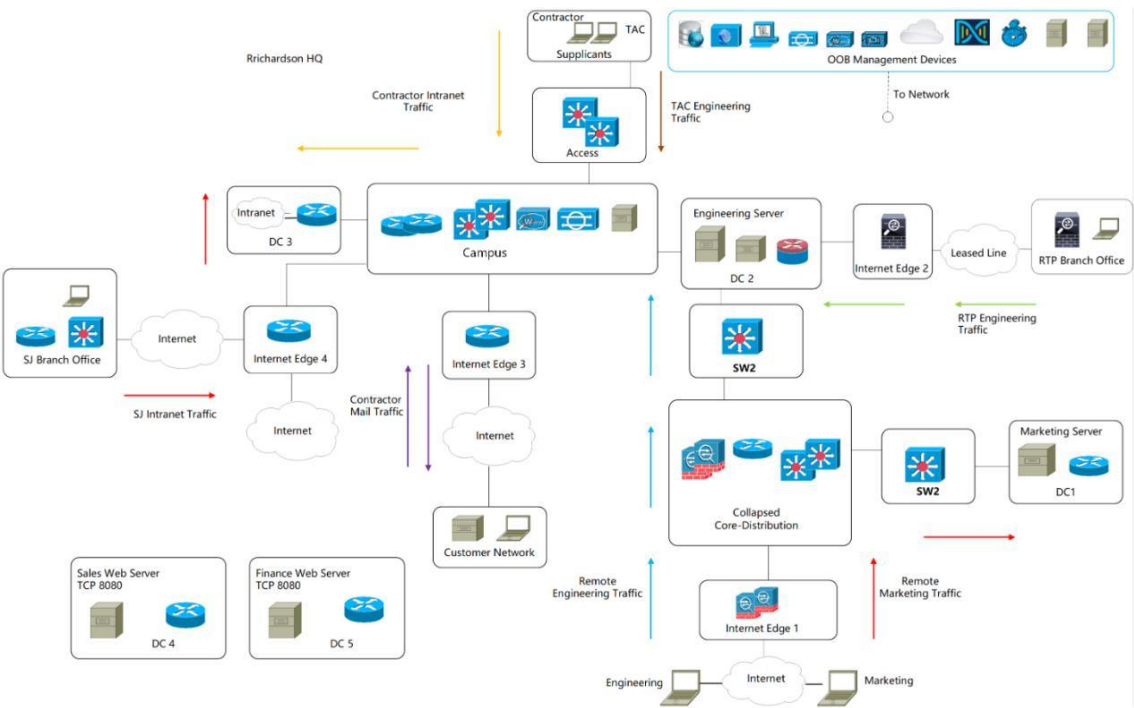


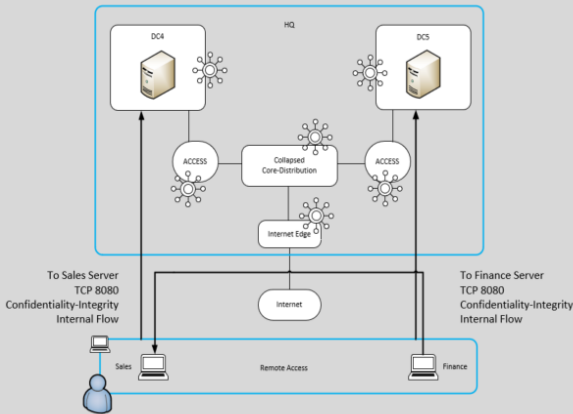
Design Resources:

>Diagrams

>Network Topology



>Diagram: Traffic Requirements and Network Analysis Outcome



HLD Categories	Attack Surface	Security Capabilities	Attack Surface	Threats	Cisco Protects
Network Security Segmentation Centralized User Information Dynamic Access Policy Address Obscurity Threat Monitorin Threat Mitigation	Human	Identity	Human		ISE
Endpoint Security Posture Assessment DNS Protection	Device	Posture Assessment Client-based Security	Device		
Network Management DNS DHCP NTP	Network	Firewall TrustSec Intrusion Detection Prevention	Network		
Network Availability Failover	Application	Web Security Email Security Malware Sandbox Server-Based Security			
Network Reachability Routing Protocol					

>Documents

>Introduction

The CTO of the PacketPiper Systems is asking for a remote access VPN solution that must be able to secure business traffic flows and provide asset compliance through which remote traffic will be originated. The design must be able to secure traffic flows from Sales and Finance employees when they remotely access organization web services at TCP port 8080 in Data Center 4 and Data Center 5 respectively. The web servers of the Sales and Finance organizations are hosted in newly developed data centers at company's HQ. The remote connection will be established by the Sales and Finance employees using company-provided PCs.

you have been hired as a Cisco consulting engineer by the customer to assist in the design, implementation, and validation of the solution.

>Network Information

Two branch offices are connected to company's HQ in Richardson.

The SJ branch office is connected to the HQ using L2VPN accross the Internet.

Branch office users utilize services that are hosted in Data Center 3 at the HQ. Traffic that originated from the branch office and is destined to Data Center 3 is subject to access policies when it moves through the HQ compus. An access switch in the branch office is responsible for on-boarding the clients.

The RTP branch office is connected to the HQ using site-to-site VPN across hte leased line with Cisco Firepower Threat Defense at the tunnel tail and head ends. Branch office users utilize service that are hosted in Data Center 2 at the HQ.

Marketing and engineering remote users use clientless VPNs to establish secure connections to the HQ. The Internet edge layer has ASAs configured in high availability mode. The Internet edge then connects to collapsed Core-distribution layer that has ASAs configured for high throughput. Marketing users utilize services that are hosted in Data Center 1 and Engineering usres utilize services that are hosted in Data Center 2 at the HQ.

The access layer at HQ provides client on-boarding for the contractors using MAB and for the TAC engineers using 802.1x. Contractors utilize services that are hosted in Data Center 3 and TAC engineers utilize services that are hosted in Data Center 2 at the HQ. Traffic that originates from the contractors and TAC engineers and is destined to their respective data centes is subject to access policies when it moves through the HQ compus. Zone-Based policy Firewall is deployed in Data Center 2 for traffic inspection orginated from TAC engineers.

The management domain host the company's security appliances, such as, Cisco Identity Engine (ISE), Cisco web Appliance (WSA), and Cisco Email Security Appliance (ESA), Firepower Management Center (FMC), Cisco Next-Generation Intrusion Prevention System (NGIPS), Cisco FireAMP Cloud, Cisco Digital Network Architecture Center (DNAC), and Cisco Stealthwatch Management Console (SMC). The management domain also hosts the company's Active Domain, DNS Server, Syslog Server, and master NTP source.

Cisco ISE provides user identity services and is responsible for segmentation using Cisco TrustSec. ISE also enables RTC using Adaptive Network Control (ANC) with Cisco FMC and Stealthwatch that use pxGrid to communicate with ISE.

Cisco FMC provides the management console for FTDs and NGIPS. Cisco FMC also monitors indicators of compromise (IOCs) of on-boarded clients via its communication with Cisco FireAMP cloud. Cisco FMC retrieves SGT information from ISE using PxGrid to implement access policies and it also probes user presence in the company's Active Directory for passive authentication of on-boarded clients.

Cisco Web Security Appliance is responsible for web security services and user authentication using the company's Active Domain.

EIGRP and OSPF are deployed as authenticated routing protocols across different architecture layer of the network.

>Email

>Email: Desing Recommendation

Engineer: Based on the business requirements to protect the traffic flows, my recommendations are as follows for the remote access VPN solution:

1. **Connections must be highly available.**
2. **Any changes to the reachability of the servers must be dynamically learned and authenticated. That said, we need a static routing mechanism at the traffic tunnel terminal point.**
3. **Remote connection must allow access only to specific network services.**
4. **Traffic segmentation must be implemented for the traffic flows at HQ.**
5. **Remote users must be authenticated by a centralized identity source.**
6. **Access polices for remote users must be dynamic. --CoA.**
7. **Real address of web servers must be hidden from the outside access.**
8. **Network devices that are part of the design must use existing management domain for OOB access.**
9. **Network devices that are part of the design must by synchronized with existing network NTP source.**
10. **DNS protection must be incorporated in the design.**
11. **Traffic flow monitoring must be incorporated in the design for threat detection.**
12. **Threat mitigation must be incorporated in the design as part of rapid threat containment.**

Take a look at it and let me know if you have any question or concerns. I will set up a kickoff meeting with your operations team to formally start the project.

Customer: Thanks for the recommendations. They seem like a good starting point and will definitely server well to scope the future conversations. I will look for the meeting invite.

>Email: Desing Categories

Customer: Do you have first pass on major design components that you can categorize after the kickoff call? I want to communicate this information to our operations team as a progress update and to accommodate any of their feedback.

Engineer: Yes, we can compile this information. Based on our discussion during the kickoff call, we have categorized desing components in the areas of network availability, network reachability, network management, and supplicant security. Will keep you posted with more information as we move forward in these areas of design.

>Email: Security Capabilities

Engineer: In the last call with the operations team, there was a dicussion on various attack surfaces in the proposed design that must be protected. Now that we have a decent picture of what the design would like for the remote access VPN solution, we can identify the attack surface accross the different design architecture layers in the proposed solution. The attack surfaces that we need to be concerned about are Human, Device, Network, and Application. I will keep you posted with more information on these attack surface locations in the proposed solution and their corresponding security capabilities for protection. Thanks!

Customer: Thanks for the follow-up on this pending item. Appreciate it.

>Email: Attack Surfaces

Customer: In the last email you touched on the locations of various attack surfaces in the proposed network design. I had an internal call with the operations team and they are very interested to know those points in network. They want to leverage that information to identity any potecial security holes in the existing network to insure end-to-end security. Let me know if this is something that you can share at this point.

Engineer: Most certainly. I will be sending a formal document with information on attack surfaces to PIN mapping very soon.

>Email: Capabilities Classification

Engineer: We are at the phase of design where we have the list of security capabilities that can provide desired security controls for the business-critical traffic flows. The next logical progression of the design process is to classify those security capabilities that would help us group networking elements that would be included in those categories. The categorization is useful to visualize the business-critical traffic that is traversing through certain networking elements and which type of security capabilities they must provide. This in turn helps us understand the potecial point of failure in the design, and helps us scope finances for the desired solution. We are working with three categories in the proposed design, foundational, Access, and Business. I will set up a follow-up call with the operations team to further discuss these categories and their corresponding security capabilities for the porposed design.

Customer: Greatly appreciate you sharing this information. Looking forward to getting the call invite.

>Email: Cisco Products

Engineer: Now that we have identified attack surfaces in the purposed designed, we can start deciding on the security products that will provide the required security capabilities to protect the traffic flows. I will set up another call with the operations team to discuss this next phase.

Customer: Glad to know that. When we have this information, we can start working toward the design business proposal.

>Email: Progress Updates 1

Engineer: I Would like to let you know that at this point we have completed the initial phase of the design in which we have covered requirements to protect the traffic flows and existing network analysis so that we can perform seamless integration of the solution. I have attached its outcome, with hopefully will help you visualize where we are at this point. The next phase will be to design, implement, and validate technical control pertaining to perimeter security and high availability components of the solution.

I Will keep you posted as we make progress, but feel free to let me know if you have any question or concerns at this time.

Customer: Glad to know that and thanks for sharing the summarized information. My team and I are looking forward to working with you on the next phase of design.

Customer: We had an our internal discussion with the operations team yesterday about ASAs connected to DC 4 and 5, and we want to incorporate these design components:

1. **Firewalls must act as a routed hop.**
2. **Each firewall must be divided into two logical firewalls.**
3. **Each physical interface on the ASA must be shared between two logical firewalls.**
4. **Each firewall must serve three security zones.**
5. **There must be a dedicated context for remote root access in each ASA.**

Engineer: Thanks for sharing this information. We will validate and include them in the design.

Engineer: After further review, we recommend these configuration components:

1. **ASA1 should be primary in the pair.**
2. **Sales traffic should be routed through ASA1 context C1.**
3. **Finance traffic should be routed through ASA2 context C2.**
4. **Each context should not have more than five SSH, ASDM, and Telnet connections.**

Let me know if you have any question or concerns before we finalize them in the design.

Customer: I have checked with the operations team and they are OK with the recommendations.

>Email: Translation on ASAs

Customer: After another call with the operations team last week, we would like to add these translation requirements for the Sales server that is hosted in the DMZ:

1. Consistent mapping between real and mapped address.
2. Any side must be able to initiate connection.
3. NAT is configured as a parameter for network object.
4. Context must have only relevant traffic stream translation.

Engineer: Thanks for the information. Again, we will validate and include them in the design and keep you posted.

Engineer: While we are working on the request for Sales server translation, do you have any specific requirements for Finance server translation too?

Customer: Glad that you have asked. The operations team makes these requests for the Finance server, very similar to Sales, just minor variations.

1. Consistent mapping between real and mapped address.
2. Any side must be able to initiate connection.
3. Network object is the parameter of NAT configuration.
4. Context must have only relevant traffic stream translation.

Engineer: Thanks for sending this over, we will validate and integrate.

>Email: ASAs Hardware

Customer: We had some internal discussion about understanding ASA1 and ASA2

requirements for the current traffic needs and also for future scalability. We are expecting a peak traffic burst in the range of 1 Gbps and expecting about 200 thousand simultaneous connections at the rate of around 20 thousand connections per second during the peak hours. We do need high availability of the solution and expect ASAs to provide inspection capability of traffic flows using HTTP, FTP, SMTP, and DNS protocols. Let us know which ASA model you would recommend. Also, while you are looking into it, can you please share the CCO link that has ASA models specifications?

Engineer: Sure. Let us investigate and get back to you.

<https://www.cisco.com/c/en/us/products/collateral/security/asa-firepower-services/datasheet-c78-742475.html>

Engineer: While we are working to recommend ASA1 and ASA2 platform, I was wondering do you have any specific requirements for ASA1v and ASA11v for the platform selection?

Customer: I was about to send an email to address the requirements for ASA1v and ASA11v. We are expecting a peak traffic burst in the range of 0.2 to 0.25 Gbps so ASAs must be able to support it. The ASAs must be able to sustain 45 to 50 thousand simultaneous remote connections at any given time, and in terms of rate, expecting 4000 to 5000 connections per second during the peak hours. These connection requirements must be coupled with high availability of the solution and must be able to perform deep inspection of traffic using HTTP, FTP, SMTP, and DNS protocols.

>Email: ASA1v-ASA11v HA

Customer: There is on piece of information that I have not sent your way, which are ASA1v and ASA11v high availability desing requirements compiled by the operations team. Here they are:

1. **Firewalls must act as a routed hop.**
2. **Only one firewall must pass traffic for both Sales and Finance organization at any given time.**
3. **Each firewall must have two security zones.**
4. **The firewall interface in the Core-Disctribution layer must have highest security.**
5. **ASA1v must be active in the peer when no failure is detected.**
6. **DNS queries must be sent across the VPN tunnel.**
7. **Only traffic between Sales and Finance organizations and the traffic destined to Sales and Finance servers must be encrypted.**

Engineer: No problem, it's not too late. We will incorporate these requirements in the design and keep you posted if we run into any issues.

Design Questions:

1. Ex:

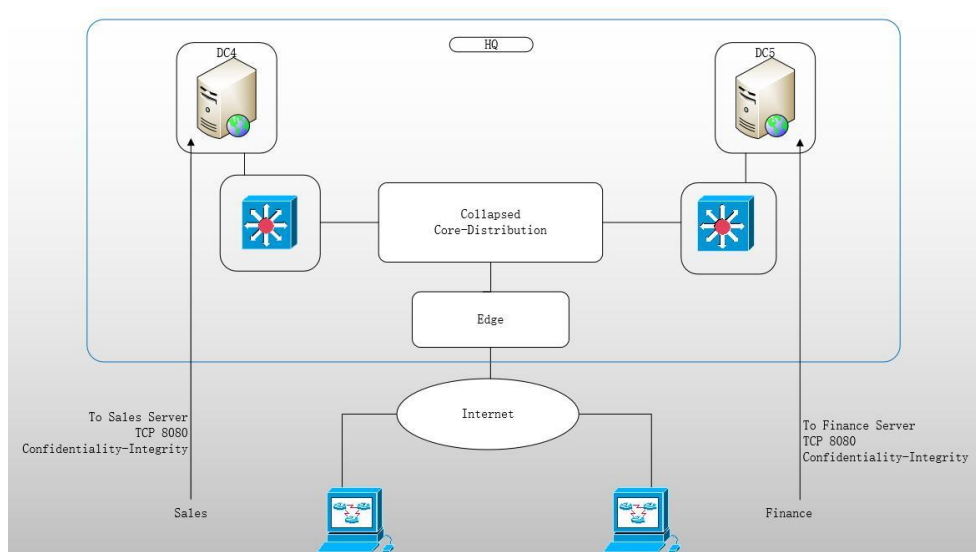
Refer to the new resource(s) available.

2. Which for statements correctly represent Sales and Finance Organization traffic flows? (choose four).

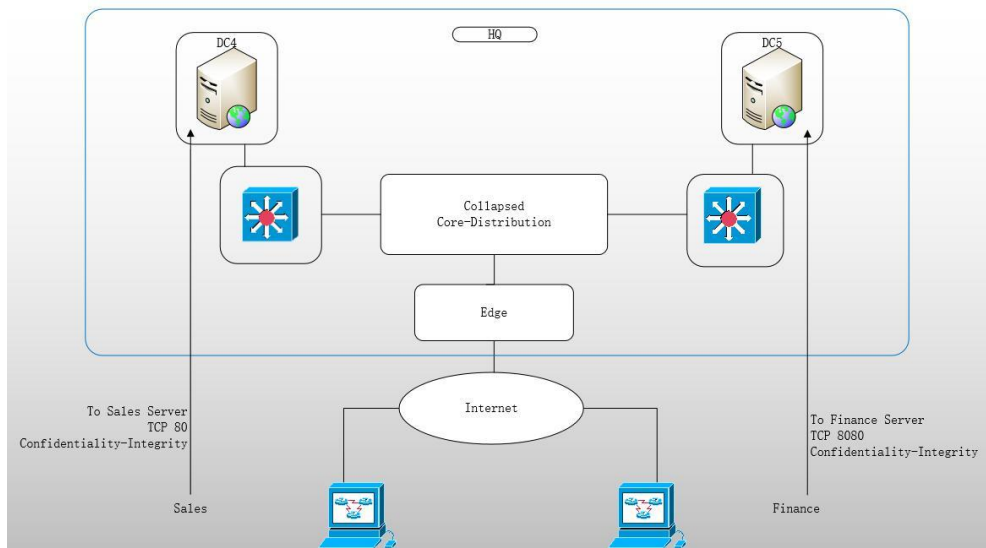
- ☐ The Sales and Finance web service ports are at UDP 8080.
- ☒ The Sales and Finance web service port is at TCP 8080.
- ☒ DC4 is hosting Sales web services and DC5 is hosting Finance web services.
- ☐ DC5 is hosting Sales web services and DC4 is hosting Finance web services.
- ☐ Sales and Finance traffic requires Only confidentiality.
- ☐ Sales and Finance traffic requires Only integrity.
- ☒ Sales and Finance traffic requires confidentiality and integrity.
- ☒ Sales traffic is destined for DC4 and Finance traffic is destined for DC5.
- ☐ Sales traffic is destined for DC5 and Finance traffic is destined for DC4.

3. Which architecture represents the correct flow for the design?

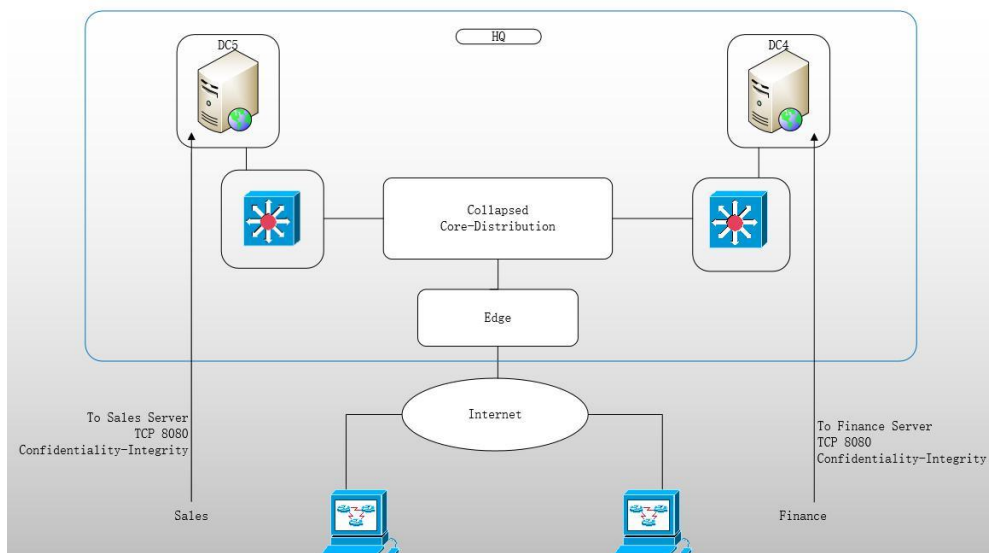
- ☒ Architecture 1



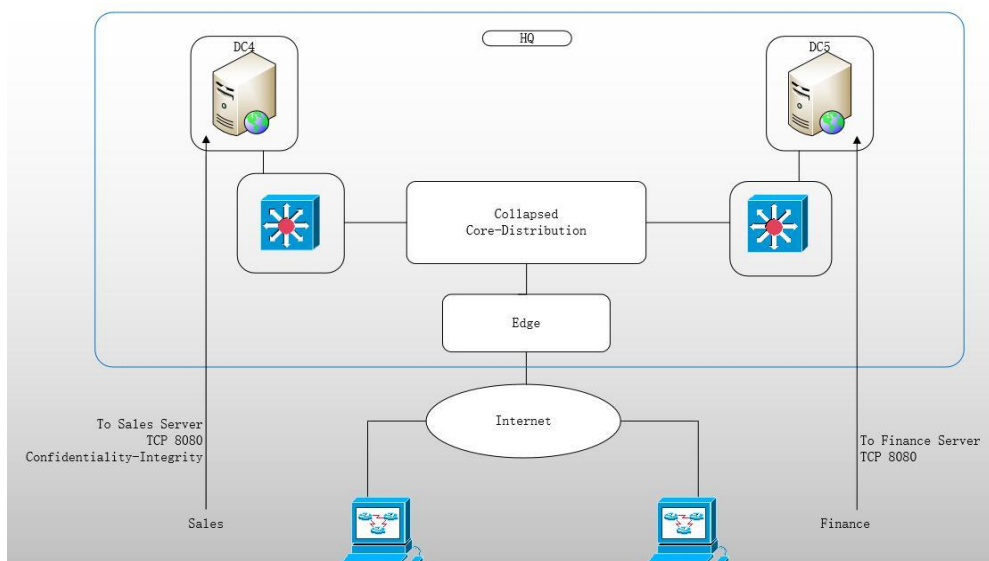
- ☐ Architecture 2



○ Architecture 3



○ Architecture 4



4. Which technology components map to the design categories to realize the initial logical solution? (choose all that apply)

Technology Components								
Design	DNS	DHCP	Routing Protocol	Posture Assessment	Failover	Centralized User Database	NTP	DNS Protection
Endpoint Security	☐	☐	☐	☒	☐	☐	☐	☒
Network Management	☒	☒	☐	☐	☐	☐	☒	☐
Network Availability	☐	☐	☐	☐	☒	☐	☐	☐
Network Reachability	☐	☐	☒	☐	☐	☐	☐	☐

5. Which security functions map to the network elements to realize the initial logical solution? (Choose all the apply.)

Security Functions						
Network Elements	Network Segmentation	Address Obscurity	Request Services	Network Convergence	Host Credentials	Forward Access Request
Identity Source	☐	☐	☐	☐	☒	☐
Suplicant	☐	☐	☒	☐	☐	☐
Authenticator	☐	☐	☐	☐	☐	☒
Firewall	☐	☒	☐	☐	☐	☐
Dynamic routing	☐	☐	☐	☒	☐	☐
Authentication Server	☒	☐	☐	☐	☐	☐

Refer to the new resource(s) available.

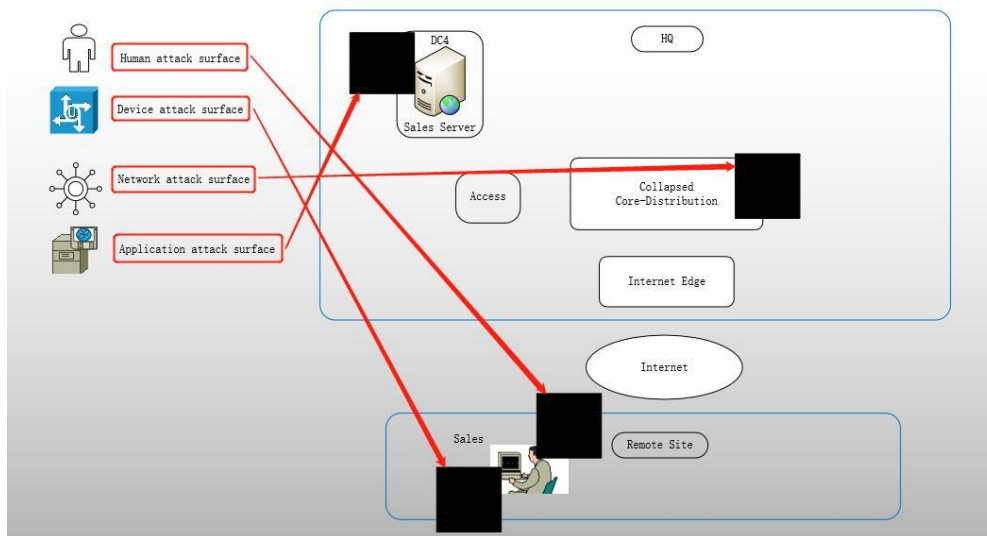
6. Which statement correctly defines Sales and Finance traffic flows and its corresponding design functional control?
- ☐ The traffic is an intranet flow with design that provides secure local access functional control with no high availability
 - ☐ The traffic is an internet flow with design that provides secure local access functional control with no high availability
 - ☐ The traffic is an external flow with design that provides secure local access functional control with no high availability
 - ☐ The traffic is an internal flow with design that provides non-secure remote access functional control with high availability
 - ☒ The traffic is an internal flow with design that provides secure remote access functional control with high availability

- The traffic is from a customer with design that provides secure remote access functional control with high availability
- The traffic is from a vender with design that provides secure local access functional control with high availability

7. Which security capabilities map to the attack surfaces to scope cisco security products for the solution? (Choose all that apply.)

Security Capabilities											
Attack surfaces	Identity	Client-Based Security	Posture Assessment	Fire wall	Trust Sec	Intrusion Detection	Email Security	Web Security	Malware Sandbox	Application Visibility control	Server-Based Security
Human attack surface	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Device attack surface	☐	☒	☒	☐	☐	☐	☐	☐	☐	☐	☐
Network attack surface	☐	☐	☐	☒	☒	☒	☐	☐	☐	☐	☐
Application attack surface	☐	☐	☐	☐	☐	☐	☒	☒	☒	☐	☒

8. Which places in the network (PINs) map to the attack surfaces to protect the traffic flows of the Sales and Finance organizations?



Refer to the new resources(s) available.

9. Which attack surface in the design, When compromised, results in the by-pass of downstream technical controls?

- Core
- Access
- ☒ Human
- Application
- Device

- Distribution
- Network

Refer to the new resource(s) available.

10. Which identity-based control, When incorporated into the design, mitigates threats that originate from the remote company assets?

- Providing contextual knowledge of threats
- Controlling viruses that are compromising the systems
- Performing device compliance
- Stopping redirection to malicious web sites
- Assigning role-based access controls
- Identifying attack using the known signatures

Refer to the new resource(s) available.

11. Which two attack surfaces in the design can be used to define the traffic baseline for anomaly detection? (Choose two.)

- Access
- Human
- Device
- Network
- Application
- Core
- Distribution

Refer to the new resource(s) available.

12. Which security capabilities map to the security categories to realize the initial solution? (Choose all that apply.)

Security Capabilities						
Security Categories	DNS Security	Posture Assessment	Intrusion Prevention System	Firewall	Identity Engine	Web Security
Foudational	☐	☐	☒	☒	☐	☐
Access	☐	☒	☐	☐	☒	☐
Business (may or may not be internal)	☒	☐	☐	☐	☐	☒

Refer to the new resource(s) available.

13. Which Cisco products map to the attack surfaces to realize the initial logical design?
(Choose all that apply.)

Cisco Products								
Attack Surfaces	Cisco Routers	Cisco ISE	Cisco ASA	Cisco NGIPS	Cisco AMP	Cisco Anyconnect	Cisco Switches	Cisco Umbrella
Human	☐	☒	☐	☐	☐	☐	☐	☐
Device	☐	☐	☐	☐	☒	☒	☐	☒
Network	☒	☐	☒	☒	☐	☐	☒	☐

Refer to the new resource(s) available.

14. Choose the correct options to develop a valid ASA high availability configuration for the solution.

ASA1 and ASA2 configured in (Routed/Transparent/Hybrid) mode with (single context C1 and nested context C2/multiple context C1 and nested context C2/multiple contexts C1 and C2) defined. Each ASA has a dedicated (system/admin/non-admin) context and three (subinterfaces/trunk interfaces/template interfaces) configured for each of the physical interface on the ASA.

Refer to the new resource(s) available.

15. Choose the correct options to develop a valid ASA high availability configuration for the solution.

ASA2 configured as (primary/secondary/standby) unit in the high availability setup. The traffic for the Sales organization routed through (ASA1-C1/ASA1-C2/ASA2-C2) context and traffic for the Finance organization routed through (ASA1-C1/ASA1-C2/ASA2-C2) context. For context resources the (no/default/custom) class is applied.

Refer to the new resource(s) available.

16. Choose the correct options to develop valid NAT configuration for the solution.

In the (Internet Edge/Access/Core-Distribution/Datacenter) layer deploy (Static NAT/Dynamic NAT/PAT/Identity NAT) using (Network Object/PAT Object/ Twice NAT/ Twice PAT) for (Sales server/Finance server/Engineering server/Marketing server) from (Inside/Outside/DMZ) to (Inside/Outside/DMZ) routed through (ASA1-C1/ASA2-C2/C1 and C1)

Refer to the new resource(s) available.

17. Choose the correct options to develop a valid NAT configuration for the solution.

In the (Internet Edge/Access/Core-Distribution/Datacenter) layer deploy (Static NAT/Dynamic NAT/PAT/Identity NAT) using (Twice NAT/Twice PAT/Network Object/PAT Object) for (Sales server/Finance server/Engineering server/Marketing server) from (Inside/Outside/DMZ) to (Inside/Outside/DMZ) routed through (ASA1-C1/ASA2-C2/C1 and C2)

Refer to the new resource(s) available.

18. Which model is the best fit to deploy ASA1 and ASA2?

- ☐ 5506
- ☐ 5508
- ☒ 5516
- ☐ 5525
- ☐ 5545
- ☐ 5555

Refer to the new resource(s) available.

19. Which model is the best fit to deploy ASA1v and ASA11v?

- ☒ 5506
- ☐ 5508
- ☐ 5516
- ☐ 5525
- ☐ 5545
- ☐ 5555

Refer to the new resource(s) available.

20. Which four configuration components enable a valid ASA high availability configuration? (Choose four.)

- ☐ ASA1v and ASA11v configured as secondary unit in the failover pair.
- ☒ ASA1v-ASA11v have na inside route for DNS reachability.
- ☐ ASA1v-ASA11v have an DMZ route for DNS reachability.
- ☒ ASA1v-ASA11v configured in routed mode as Active-Standby failover.
- ☐ ASA1v-ASA11v configured in transparent mode as Active-Standby failover.
- ☐ ASA11v configured as primary unit in the failover pair.
- ☒ ASA11v configured as secondary unit in the failover pair.
- ☐ DNS queries resolved locally on Remote PCs for protection Against DNS attack.
- ☐ DNS queries performing through the ISP assigned DNS servers.
- ☐ Split-tunneling enabled only for non-interested traffic.
- ☒ Split-tunneling enabled only for interested traffic.
- ☐ ASA1v-ASA11v have na outside route for DNS reachability.
- ☐ ASA1v-ASA11v configured in transparente mode as Active-Active failover.
- ☐ ASA1v-ASA11v configured in routed mode as Active-Active failover.
- ☐ ASA1v and ASA11v configured as primary units in the failover pair.

Refer to the new resource(s) available.

21. Which config line correctly maps to its functionality in the ASA1v configuration provided to the customer?

failover	Active and standby addresses should be in the same subnet.
failover interface ip F0 5.2.9.1 255.255.255.0 standby 5.2.9.2	It enables failover.
failover lan interface F0 GigabitEthernet0/2	It keeps the configuration in-sync.
failover lan unit primary	It monitors the health of the interface that faces the Internet.
failover link F0 GigabitEthernet0/2	It prevents traffic interruption at failover.
monitor-interface inside	It triggers failover when a higher security link is down.
monitor-interface outside	The other unit has standby addresses.

Refer to the new resource(s) available.

22. Which config line correctly maps to its functionality in the ASA1 configuration provided to the customer?

allocate-interface Management0/0	Delay in seconds for connections replication.
interface GigabitEthernet0/0.1	It allows OOB management.
join-failover-group 1	It drops the sixth TCP session at port 23.
join-failover-group 2	It provides connectivity to the campus.
limit-resource Telnet 5	It routes traffic for the Finance organization.
preempt 100	It routes traffic for the Sales organization.

Refer to the new resource(s) available.

23. Which config line correctly maps to its functionality in the ASA1 configuration provided to the customer?

allocate-interface GigabitEthernet0/0.2 inside_c2	ASA2 interface in this group actively pass traffic.
context C1	Creates a virtual firewall.
failover group 2	Interface has the active address on ASA2.
failover lan unit primary	It designates ASA2 as a secondary unit.
interface GigabitEthernet0/3	It facilitates replication of the configuration.
interface GigabitEthernet0/4	It facilitates replication of the connections.

Refer to the new resource(s) available.

24. Which config line correctly maps to its functionality in the ASA2 configuration provided to the customer?



Refer to the new resource(s) available.

25. Which Technologies When configured on ASA1 and ASA2 will provide functional controls to fulfill the design requirements?

Functional Controls						
Technologies	Real address of the Sales and Finance servers not visible in Internet Edge	ACL with source and destination address replaced by SGTs	Dynamic and static SGTs pushed by ISE	Bandwidth and delay to calculate the shortest routing path	Translated addresses of the Sales and Finance servers reachable using routing protocol	RADIUS server that provides NAD authentication
EIGRP	☐	☐	☐	☒	☐	☐
Redistribution	☐	☐	☐	☐	☒	☐
Route Filtering	☒	☐	☐	☐	☐	☐
SXP	☐	☐	☒	☐	☐	☐
ISE	☐	☐	☐	☐	☐	☒
SGACL	☐	☒	☐	☐	☐	☐

Refer to the new resource(s) available.

26. Which commands When executed on ASA1-ASA2 and ASA1v-ASA11, show the outputs that have been provided to the customer in the design validation document? (Choose all that apply.)

Validation Outputs						
Show Commands	Interface Security Level	Interface Monitoring	Client Username	Local Mode	ISE IP Address	Client IP Address
Sh nameif	☒	☐	☐	☐	☐	☐
Sh failover	☐	☒	☐	☐	☐	☐
Show vpn-sessiondb anyconnect	☐	☐	☒	☐	☐	☒

Show cts sxp connections	☐	☐	☐	☒	☒	☐
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Refer to the new resource(s) available.

27. Which technologies must be configured on these ASAs to provide functional controls to fulfill the design requirements? (Choose all that apply.)

ASAs				
Technologies	ASA1v	ASA11v	ASA1	ASA2
Virtual Firewalls	☐	☐	☒	☒
DNS	☒	☒	☐	☐
EIGRP	☐	☐	☒	☒
DACL	☒	☒	☐	☐
VPN	☒	☒	☐	☐
SXP	☒	☒	☒	☒

Refer to the new resource(s) available.

28. Which technologies must be configured on these ASAs to provide functional controls to fulfill the design requirements? (Choose all that apply.)

ASAs						
Technologies	ASA1v	ASA11v	ASA1-C1	ASA1-C2	ASA2-C1	ASA2-C2
AnyConnect	☒	☒	☐	☐	☐	☐
RADIUS	☒	☒	☐	☐	☐	☐
NAT	☐	☐	☒	☐	☐	☒
SGACL	☒	☒	☒	☐	☐	☒

Refer to the new resource(s) available.

29. Which config line correctly maps to its functionality in the ASA1v configuration provided to the customer?

aaa-server ccie (mgmt) host 150.1.7.218	ISE receives Sales and Finance session accounting updates every 60 minutes.
aaa-server ccie protocol radius	It identifies to which group ISE belongs.
authentication-server-group ccie	Listen for policy updates from ISE as part of CoA.
dynamic-authorization	The ISE with the session tunnel-group attributes.
interim-accounting-update periodic 1	The maximum allowed length is 64 characters.
key cisco	Use the management interface for communication with ISE.

Refer to the new resource(s) available.

30. As part of the validation document, which information on the ASAs can be validated by using the show vpn-sessiondb anyconnect command?

Information about the (failed/established/failed and established) session showing (static/assigned/static and assigned) client (IP address/MAC address/tunnel address) along with the session (public IP address/private IP address/public and private IP address) and (DACL/SXP/SGT).

Refer to the new resource(s) available.

31. Which technologies When configured on ASA1v and ASA11v provide functional controls to fulfill the design requirements?

Funcional Controls								
Technologies	External datastore for Sales and Finance Authentication	Data confidentiality implemented using AES 256	Learning SGT assigned to endpoints in the network	Session should remain active for 2 Hours without any activity	ISE pushing access-control policy for the traffic flows	ISE local datastore for Sales and Finance authentication	Anyconnect clients and NAD authentication using RADIUS	ACL with source and destination address replaced by SGTs
IKEv2 encryption proposal	☐	☒	☐	☐	☐	☐	☐	☐
DACL	☐	☐	☐	☐	☒	☐	☐	☐
VPN idle-timeout	☐	☐	☐	☒	☐	☐	☐	☐
SXP	☐	☐	☒	☐	☐	☐	☐	☐
SGACL	☐	☐	☐	☐	☐	☐	☐	☒
ISE	☐	☐	☐	☐	☐	☐	☒	☐
Backup authentication	☐	☐	☐	☐	☐	☒	☐	☐
Active directory	☒	☐	☐	☐	☐	☐	☐	☐

Refer to the new resource(s) available.

32. Which six configuration componentes implemente the feature that the customer has recently requested? (Choose six.)

- ☐ Quarantine policy for Windows on Cisco FMC
- ☒ Quarantine policy for Windows on Cisco AMP Cloud
- ☒ On-premises installation of Cisco AMP Cloud
- ☒ On-premises installation of Cisco Umbrella
- ☐ Correlation policy on NGIPS for IOC
- ☒ Correlation policy on Cisco FMC for IOC

- NGIPS and Cisco AMP Cloud integration with FMC
- Cisco Stealthwatch and AMP Cloud integration with FMC
- In-line NGIPS installation between ASA1v-ASA11v and ASA1-ASA2
- In-line Cisco FMC installation between ASA1v-ASA11v and ASA1-ASA2
- AMP connector installation from ASA1v on PCs
- AMP connector installation from Cisco AMP Cloud on PCs

Refer to the new resource(s) available.

33. Which eight configuration components implement the feature that the customer has recently requested? (Choose eight.)

- Redirect ACL on ASA
- Redirect ACL on ISE
- Compliant ACL to deny access to ISE and DNS
- Compliant ACL to permit access to ISE and DNS
- Unknown posture profile tied with redirect ACL on ISE
- Microsoft patch presence defined as posture remediation action on ISE
- Complaint and non-compliant ACLs on ASA
- Compliant and non-compliant ACLs on ISE
- Remediation window configured under AnyConnect configuration file on ISE
- AnyConnect configuration file configured under Posture Profile on ISE
- Remediation action as messaging to end user on compliance
- Remediation action as messaging to end user on non-compliance
- Compliant and non-compliant posture authorization profiles on ASA
- Compliant and non-compliant posture authorization profiles on ISE
- Unknown posture authorization profile tied with redirect ACL name on ISE
- Non-compliant ACL to deny access to ISE and DNS
- Non-compliant ACL to deny access to remediation server
- Microsoft patch presence defined as posture requirement condition on ISE